

GTR70E WYVERN (Widespread Yield Variable Engagement Rapid-response Neutralizer): Engineering Design and Experimental Validation of Closed-Loop Thrust-Vector Control, Hybrid Passive-Active Stability, and Zoned Additive-Manufacturing Materials in a Subscale Single Stage Solid-Fuelled Prototype Rocket Demonstrator

Student Information

Student Names: Swaroop Sahoo, Chris Liu, Allison Hong

Project Title: GTR70E WYVERN (Widespread Yield Variable Engagement Rapid-response Neutralizer): Engineering Design and Experimental Validation of Closed-Loop Thrust-Vector Control, Hybrid Passive-Active Stability, and Zoned Additive-Manufacturing Materials in a Subscale Single Stage Solid-Fuelled Prototype Rocket Demonstrator

Research Pathway: Engineering/Innovation Design

Intended Majors:

- *Swaroop Sahoo:* Electrical Engineering with Aerospace Engineering Minor
- *Chris Liu:* Aerospace Engineering with Business Minor
- *Allison Hong:* Mechanical Engineering with Aerospace Engineering Minor

Abstract

The GTR70E WYVERN vehicle is a 70 mm-diameter, single-stage, solid-fuelled, additively-manufactured prototype rocket demonstrator developed to provide quantitative, ground-based and flight-validated data on closed-loop thrust-vector control (TVC) actuation, hybrid passive/active stability architectures, and zone-specific additive-manufacturing material selection. The vehicle is powered by a single Estes F15-4 solid motor and relies on four fixed fins sized near the minimum statically-stable margin to survive launch rail departure and the motor's ignition transient, after which authority is transferred to a closed-loop proportional-integral-derivative thrust-vector-control system operating on the smooth portion of the thrust curve. Two candidate TVC actuation schemes, a tri-solenoid magnetic gimbal and a servo-actuated gimbal, are characterized and directly compared on two physically separate but instrumentally identical three-axis thrust-vector load balances prior to flight, isolating actuator dynamics from flight-to-flight variability. Fin aerofoil selection and stability-model calibration draw on a dedicated bench wind tunnel and a companion two-dimensional vortex-panel computational solver, cross-checked against flight telemetry, and the zoned-materials study additionally screens filament performance under real motor-plume conditions on the static-fire stand. All flight avionics, sensor fusion, and control-law execution are consolidated onto a single Raspberry Pi Pico 2 W dual-core microcontroller carried on a hand-assembled prototyping board, with one core dedicated exclusively to deterministic 500 Hz control execution and the second core

handling data logging to a microSD card, eliminating the blocking-input/output risk inherent to single-threaded flight computer architectures. Flight telemetry is logged to card as the data of record, and the module's onboard radio is used for bench telemetry on the ground test stand. Structural members are fabricated by fused deposition modeling using a three-way thermally-zoned material strategy: foamed acrylonitrile styrene acrylate forms the upper body tube housing the avionics, where there is neither motor heat nor ejection gas; carbon-filled polyethylene terephthalate glycol forms the lower body, the bulkhead joint, and the fins, which carry the ejection-gas path and the fin aerodynamic load; and fire-retardant polycarbonate forms the thrust-vector-control assembly proper, the zone closest to sustained motor-plume heating. Fin aerodynamics are characterized analytically by the Barrowman (1967) normal-force and center-of-pressure method embedded in the program's fourth-order Runge–Kutta trajectory suite, and validated against recovered flight telemetry, from which an in-situ drag coefficient and static margin are reconstructed. All flights are conducted under Federal Aviation Administration Class 1 model rocket provisions, requiring no airworthiness waiver. Pre-flight Monte Carlo and validation simulation predict an apogee near 409 feet with a positive thrust-to-weight ratio throughout the burn and a closed-loop pitch deviation under three degrees across the modeled atmospheric envelope. All design files, firmware, simulation code, and flight and ground-test datasets will be released publicly through a version-controlled, openly accessible repository and presented in summary form within this paper upon completion of the program.

Introduction

Background and Motivation

Active flight control in small-scale, hobbyist-accessible rocketry sits at the intersection of aerodynamics, embedded real-time systems, additive-manufacturing materials science, and classical control theory, and progress on any one axis is frequently constrained by the others. Thrust-vector control, gimbaling the rocket motor's nozzle or exhaust path to produce a control moment without dedicated aerodynamic control surfaces, is the actuation method used on essentially every orbital launch vehicle, yet open, reproducible, and quantitatively validated implementations at the sounding-rocket scale remain comparatively rare in the amateur and academic literature. Existing low-cost demonstrations of vector control, such as the BPS.space *Signal* and *Echo* flight series, establish feasibility but do not typically isolate and quantify actuator-class performance independent of flight-to-flight aerodynamic and atmospheric variability.

GTR70E WYVERN addresses this gap with a deliberately simplified architecture relative to prior program iterations: a single fixed-fin airframe that is passively stable through the motor's ignition transient, a single bare-metal flight controller executing the entire sensing-and-control pipeline, and, critically, an actuator comparison that is moved off the flight vehicle and onto a repeatable ground-test apparatus. This restructuring is intended to produce a controlled, statistically tractable dataset on actuator performance, the central engineering question of the program, while retaining a flight-validation phase that demonstrates the complete closed-loop system performs as designed under real atmospheric and motor-burn conditions.

Program Lineage

The GTR70E WYVERN line has progressed through four major design iterations. The first was a 70 mm two-stage airframe with a custom flight computer, magnetic-solenoid thrust-vector-control actuation, and actively controlled fins on both stages. The second introduced an 84 mm two-stage airframe with custom avionics. The third implemented a Raspberry Pi 5-based flight computer and flew a magnetic-solenoid-versus-servo actuator comparison in flight. Each iteration established components of the aerodynamic, structural, and avionics groundwork carried forward into the current design. The current iteration consolidates these lessons into the simplest vehicle configuration that still answers the program's central control-authority question, replacing the in-flight actuator comparison with a ground-based load-balance comparison and replacing the distributed avionics stack with a single dual-core microcontroller.

Central Hypothesis

A small fixed-fin, single-stage, solid-fuelled prototype rocket demonstrator can rely on passive aerodynamic stability to survive launch-rail departure and the motor's ignition transient, then transfer control authority to a closed-loop thrust-vector-control system once the thrust curve smooths; and the relative performance of competing thrust-vector-control actuator classes, magnetic-solenoid versus servo, can be characterized quantitatively, and more economically, on a ground-based three-axis load balance prior to committing either actuator to flight.

Research Questions

The program addresses five research questions spanning actuator selection, materials, aerofoil performance, stability-model calibration, and control-loop tuning. Each is addressed through at least two independent and mutually cross-validating methods drawn from the three experimental tiers described later in this proposal.

Magnetic-Solenoid Versus Servo Actuation for Thrust-Vector Control Authority

Substituting a tri-solenoid magnetic gimbal actuator for a servo-driven gimbal actuator as the thrust-vector-control mechanism, evaluated under identical commanded-deflection inputs on two physically separate three-axis thrust-vector load balances sharing an identical flexure and data-acquisition chain so the two datasets are directly comparable, will produce measurable differences in actuation bandwidth, slew rate, step-response overshoot, steady-state deflection error, and maximum achievable gimbal angle, with the objective of determining which actuator class provides superior control authority for a given motor thrust regime.

Zoned Airframe Materials and Blast-Shield Materials Screen

Holding all fused-deposition process parameters constant across foamed acrylonitrile styrene acrylate, carbon-filled polyethylene terephthalate glycol, and fire-retardant polycarbonate, this study characterizes and compares each material's flexural stiffness and modulus under three-point bend loading, and its heat-deflection margin against the

measured engine-bay wall temperature from the static-fire campaign. The objective is to justify the zoned allocation actually flown: foamed acrylonitrile styrene acrylate for the unheated avionics-housing upper body, carbon-filled polyethylene terephthalate glycol for the lower body and fins, and fire-retardant polycarbonate for the thrust-vector-control assembly. The question the comparison answers is whether the fire-retardant grade and the carbon-filled stiffness are necessary in those zones, or whether a lighter single-material airframe would have sufficed, a decision every low-cost printed thrust-vector-control vehicle has to make and few justify with data. A second materials-focused thread under this same research question, run as a blast-shield screen rather than a mounted-vane test, fires the motor directly into a flat five-millimetre, fully-dense coupon plate of each of six candidate materials, measuring melt-through, ablation depth, and slag buildup. Materials that survive five millimetres are retested at four, three, and two millimetres until they fail, producing a failure-thickness ranking across all six. This is a materials data point rather than a flight-hardware decision, since only the servo thrust-vector-control system flies.

Fin Aerofoil Selection via Wind-Tunnel-Measured Performance

A bench wind tunnel directly measures lift, drag, and stall behaviour for the candidate fin aerofoil sections across the flown fin's angle-of-attack range, complementing an inviscid two-dimensional vortex-panel computational solver that predicts the same lift curves and surface-pressure distributions analytically. The objective is to select the fin aerofoil section on measured rather than purely modelled performance, and to characterize where the inviscid panel-method prediction diverges from measured behaviour. The panel method captures circulation-driven lift but not viscous separation, stall onset, or pressure drag, which only the tunnel run can supply.

Wind-Tunnel-Versus-Flight Calibration of Passive Stability

Sizing the four fixed fins near the minimum conventionally stable static margin by the Barrowman (1967) normal-force and center-of-pressure method, the predicted static margin, zero-lift drag coefficient, and weathercock response are compared against two independent measurements: the tunnel-measured aerofoil coefficients, and the same quantities reconstructed from recovered flight telemetry. Static margin is recovered from the observed pitch-rate response to the measured crosswind, and drag coefficient from the coast-phase deceleration between burnout and apogee. The objective is to establish how accurately a Barrowman-class analytical model, cross-checked against both a physical wind-tunnel measurement and flight telemetry, predicts the as-built vehicle's passive aerodynamics, and to confirm that the selected fin geometry holds a stable margin through the pre-control phase without an unnecessary drag or mass penalty.

Closed-Loop Control Gain Sensitivity in a Single-Controller Architecture

Operating the proportional-integral-derivative thrust-vector control loop on the single flight microcontroller under multiple candidate gain sets, including a step-response-tuned baseline and a simulation-refined gain set validated across modelled atmospheric and gust conditions, peak pitch deviation, gimbal-angle utilization, and settling behaviour will be quantified and compared across repeated flights. The objective is to determine which gain

configuration produces the most accurate and best-damped trajectory tracking on a deterministic single-controller architecture.

Project Goal

The goal of the GTR70E WYVERN program is to design, fabricate, ground-test, and flight-validate a recoverable, Federal Aviation Administration Class 1, 70 mm-diameter prototype rocket demonstrator that serves as an integrated experimental platform for quantitatively comparing thrust-vector-control actuator classes independent of flight-to-flight variability; validating a thermally-zoned additive-manufacturing material strategy, including filament performance under real motor plume conditions, against airframe mass and structural-margin objectives; selecting a fin aerofoil section on wind-tunnel-measured performance; establishing the predictive accuracy of a Barrowman-class analytical stability model against both wind-tunnel measurement and the as-built vehicle's in-situ passive aerodynamics; and quantifying control-loop gain sensitivity on a consolidated single-controller avionics architecture. Testing is conducted across three complementary tiers, computational simulation, ground-based instrumented testing, and in-situ powered flight, so that each research question is addressed through at least two independent and mutually cross-validating methods.

Vehicle Architecture

Configuration and Mass Budget

The vehicle is a 70 mm outer-diameter, single-stage airframe approximately 672 mm in length, comprising two body tubes joined at a single bulkhead, with four fixed fins providing passive stability. The forward tube houses the avionics and the recovery wadding; the aft tube carries the parachute pack forward of the thrust-vector-control bay and the motor.

Liftoff mass is approximately 720 g, of which 60 g is propellant, giving a burnout mass near 660 g. The printed structure accounts for roughly 405 g, the avionics and power system for roughly 117 g including the onboard camera, the recovery system for 70 g, and the two gimbal servos for 24 g. The loaded motor contributes 102 g. Structural masses are computed directly from the parametric solid model rather than scaled from earlier iterations; component masses are vendor or comparable-product figures, and no bench-scale weight has yet been taken of the assembled avionics stack.

The foamed acrylonitrile styrene acrylate upper and lower tubes are the largest single mass saving in the stack, and the aft tube length was set at the shortest value that preserves both the parachute pack volume and an adequate static margin. Shortening the aft body reduces pitch inertia faster than it reduces the thrust-vector control moment arm, so the shorter vehicle has measurably higher pitch authority than a longer one of the same mass, which directly serves the control-authority questions this program is built to answer.

Stability Architecture: Passive Margin With Active Handoff

Because the closed-loop thrust-vector-control system cannot be engaged instantaneously at ignition, as servo and control-loop settling combined with the high-amplitude transient of the motor's ignition spike would risk a destabilizing initial command, the vehicle must be passively stable through approximately the first half-second of flight. Four fixed fins with a root chord of 70 mm, a tip chord of 35 mm, a leading-edge sweep of 25 mm, and a span of 87 mm are sized to hold a static margin of roughly 1.1 calibers at liftoff, with the center of gravity 45 cm from the nose and the center of pressure 53 cm from the nose by the Barrowman (1967) method. Margin increases through the burn as propellant mass is consumed.

A parametric sweep of apogee against nose ballast confirmed that adding ballast to increase margin is counter-productive: each gram of ballast lowers apogee, and the smaller fin it permits never repays the dead weight. No ballast is therefore carried, and stability is achieved purely through fin sizing.

At half a second, with the motor on the smooth sustained portion of its thrust curve, the thrust-vector controller is enabled and assumes full attitude authority, commanding the gimbal to stabilize the vehicle to vertical and subsequently execute a small commanded pitch maneuver, while the fins continue to provide passive restoring moment as a stability backstop for the remainder of the powered phase.

Proposed Methodology

The program employs a three-tier experimental framework spanning computational simulation, ground-based instrumented testing, and in-situ powered flight. Each research question is addressed through at least two of these tiers to enable cross-validation. All fabrication uses hobbyist-accessible manufacturing methods and commercially available components, consistent with the program's open-source objectives.

Airframe and Materials

All primary structural components, body tubes, nose cone, fin set, and bulkhead, are fabricated by fused deposition modeling on a Bambu Lab X1C printer using the stock brass nozzle. Print parameters, including layer height, infill pattern and density, wall count, nozzle and bed temperature, and part cooling, are held constant within each material class to isolate material as the independent variable, following the parameter-control methodology of Popescu et al. (2018). Three structural candidate materials are evaluated: fire-retardant polycarbonate, allocated to the thrust-vector-control assembly, the zone closest to sustained motor-plume heating; carbon-filled polyethylene terephthalate glycol, allocated to the lower body and fins, which carry the ejection-gas path and the fin aerodynamic load; and foamed acrylonitrile styrene acrylate for the upper body tube and nose, where neither motor heat nor ejection gas is present. Flexural stiffness and modulus of all three materials are characterized under standardized three-point bend loading following the mechanical-characterization framework of Dizon et al. (2018).

The allocation is driven by thermal exposure and structural role. The foamed material is not survivable in contact with the motor's ejection gas or sustained plume heat; the carbon-filled material handles the gas path and fin loads; and the fire-retardant polycarbonate is reserved for the zone with the highest sustained thermal exposure. Engine-bay thermal performance is verified by a first-order lumped-capacitance transient model of the motor-bay wall against the motor's burn duration, and is cross-checked against a thermocouple reading taken on the wall during the static-fire campaign. First-order structural margins on the airframe exceed the motor's peak axial and control-induced bending loads by more than two orders of magnitude, confirming that wall thickness is set by printability and handling robustness rather than by flight loads.

Fin Geometry and Passive Stability

The flight fin employs a symmetric aerofoil cross-section sized to deliver the static margin identified above. Passive aerodynamics are established through three complementary channels: analytical prediction, direct wind-tunnel measurement, and flight validation.

For the wind-tunnel measurement, four candidate fin sections, two symmetric NACA four-digit profiles, a double-wedge profile, and a flat-plate reference, are evaluated on a bench wind tunnel across a half-degree-resolution angle-of-attack sweep at a representative tunnel Reynolds number, cross-checked against the representative flight Reynolds number. This is the measured leg of the fin-selection decision; the section actually flown is the one the tunnel data favours, not the inviscid prediction alone.

For the computational cross-check, a constant-strength two-dimensional vortex-panel method following the Kuethe and Chow methodology solves the inviscid flow about each candidate section, returning lift coefficient and surface pressure distribution across the same sweep used in the tunnel campaign, plus a flat-plate skin-friction viscous-drag estimate so that lift-to-drag ratio is meaningful for section selection. The method is validated against thin-airfoil theory, returning a lift-curve slope close to the theoretical ideal with the expected inviscid thickness over-prediction. Because the panel method is inviscid, it captures circulation-driven lift and pressure distribution but not viscous separation, stall onset, or pressure drag; those come from the tunnel campaign, which is exactly what the computational model is built to be checked against.

For the analytical prediction, fin normal-force slope and center of pressure are computed by the Barrowman (1967) slender-body method with a body-interference correction, and zero-lift drag is built up componentwise from skin friction over the wetted area at the flight Reynolds number, base drag, forebody pressure drag, and fin profile drag, giving the nominal drag coefficient carried by the trajectory integrator. Reference low-angle-of-attack behaviour for symmetric sections of this class is benchmarked against the thin-airfoil and low-Reynolds-number frameworks of Lissaman (1983) and Mueller and DeLaurier (2003), and against tabulated section data from Abbott and Von Doenhoff (1959). Sensitivity of the predicted margin to build tolerance is quantified by a sweep across the plausible range of as-built center-of-gravity error, and the resulting margin, apogee, and drift distributions by Monte Carlo dispersion over the atmospheric envelope. An independent implementation of

the same geometry in OpenRocket serves as a third-party cross-check on the center-of-pressure and trajectory prediction.

For flight reconstruction, two quantities are recovered from the onboard log and compared against both the analytical prediction and the tunnel measurement. Coast-phase drag coefficient is reconstructed from the deceleration between burnout and apogee, where thrust is zero and the only forces are drag and gravity, evaluated over the high-dynamic-pressure portion of the coast and averaged, with air density taken from the barometric altitude and the measured surface conditions. Static margin is reconstructed from the weathercock response: the observed steady pitch offset into the measured crosswind, together with the pitch-rate transient at rail exit, yields the aerodynamic restoring stiffness, from which the center of pressure follows given the measured center of gravity and dynamic pressure. Agreement between predicted and reconstructed values is the reported result for this research question; disagreement is itself a quantified finding about the predictive limits of a Barrowman-class model on a short, low-Reynolds-number, additively-manufactured airframe.

Flight Computer, Sensing, and Control Architecture

All flight avionics functions, attitude estimation, control-law execution, actuator commanding, and data logging, are consolidated onto a single Raspberry Pi Pico 2 W, a dual-core microcontroller module carried on a hand-assembled prototyping board mounted as an axial card in the avionics bay. This replaces the distributed multi-board avionics architecture used in earlier program iterations. An intermediate revision of this design specified a custom circular printed circuit board built around a bare packaged die; that approach was retired on schedule grounds, since a fine-pitch quad-flat no-lead package requires a four-layer board with a continuous ground plane and the layout and fabrication turnaround did not fit the fixed launch window. Control-loop capability is unchanged, using the same silicon and the same core partition, and the change substantially reduces bring-up risk by making the avionics testable one breakout at a time rather than as a single monolithic board.

The two processor cores are functionally partitioned to preserve hard real-time determinism. The first core executes the 500 Hz thrust-vector-control loop exclusively, reading the inertial measurement units, computing nozzle deflection, evaluating the control law, and commanding the gimbal servos, and is permitted no blocking operations of any kind. The second core drains a logged-data ring buffer to a microSD card over a serial peripheral interface, isolating all non-deterministic input/output latency from the control path. This division directly addresses the principal failure mode of single-threaded flight-computer architectures, in which storage input/output can transiently block control-loop execution. Flight telemetry is logged to card as the data of record; the module's onboard radio is used for bench telemetry on the ground test stand, where the same board and firmware image are reused with the bay inertial unit omitted.

Two nine-axis inertial measurement units are deployed, one on the avionics board in the upper body tube and one mounted directly on the thrust-vector gimbal in the lower body tube, connected by a cable that crosses the bulkhead on separable leads. They are

distinguished on a shared serial bus by their address-select pin. Each runs in a fusion mode combining accelerometer and gyroscope data with the magnetometer disabled, because the magnetic field generated by the adjacent gimbal servos would otherwise corrupt a magnetically-referenced heading estimate. The two units vote against each other for attitude fault detection, and because the second unit is gimbal-mounted, gimbal-relative deflection is measured directly in flight rather than only on the ground-test balances. At ejection the two body tubes separate and the gimbal unit disconnects with them by design; the firmware treats its loss after the deployment time as expected rather than as a fault, and descent attitude comes from the bay unit. Two independent barometric sensors on the same bus provide primary and backup altitude.

The entire avionics domain runs off a light two-cell lithium-polymer pack through a resettable fuse and an arming switch into a switching regulator producing a single five-volt rail. That rail feeds the microcontroller, the storage breakout, and both gimbal servos, which are comfortably oversized for the computed gimbal torque demand; the sensors run off the microcontroller's own regulator. A bulk electrolytic capacitor at the servo feed absorbs stall transients that would otherwise brown out the controller. Pack health is monitored by a resistive divider into an analog input, with firmware warning and critical thresholds set at conservative per-cell voltages. The arming switch is reached by removing the nose cone rather than through an airframe penetration.

The per-axis control law is a discrete proportional-integral-derivative controller with integral anti-windup clamping and a first-order low-pass-filtered derivative term, executed at 500 Hz with output clipped to an eight-degree gimbal deflection limit. That limit was raised from an earlier five degrees to give control-authority margin against crosswind weathercocking without adding passive fin stability, which would otherwise reduce the very disturbance the control system is built to demonstrate. The flight gain set was selected by a phase and gain margin analysis across twenty-four operating points and independently confirmed by a time-domain robust multi-wind auto-tune. It holds worst-case gust pitch deviation to under two degrees with wide gimbal headroom against the mechanical limit, while avoiding the resonance against finite servo lag observed in higher-proportional-gain configurations. The control loop is inhibited for the first half-second of flight, after which it engages to stabilize the vehicle to vertical and execute a small commanded maneuver.

Two thrust-vector-control actuator classes are evaluated using the identical control electronics, gimbal mechanism, and software control law, isolating actuator dynamics as the experimental variable: a tri-solenoid magnetic gimbal actuator and a servo-driven gimbal actuator. The comparison is conducted entirely on the ground-based three-axis thrust-vector load balances described below rather than in flight, to remove flight-to-flight aerodynamic and atmospheric variability from the actuator comparison. The flight vehicle itself carries the servo actuator, selected on the basis of the ground-comparison results.

Ground Test Program

The ground test program is structured as the program's primary experimental apparatus rather than a preliminary check, since the actuator comparison, the materials data, the

aerofoil measurement, and the motor-characterization data that feed every downstream simulation are all generated there rather than in flight. Four purpose-built stands are constructed: a servo thrust-vector stand, a physically separate magnetic thrust-vector stand, a static-fire stand handling calibration, thrust curves, and the blast-shield materials screen, and a bench wind tunnel for aerofoil measurement. The three motor-fired stands are fully printable in carbon-filled polyethylene terephthalate glycol, selected for its motor-plume thermal margin, and instrumented with strain-gauge load cells and twenty-four-bit bridge-amplifier breakouts, logged to onboard storage by a dedicated data-acquisition microcontroller independent of the flight avionics. The wind tunnel is an unpowered bench rig with its own instrumentation.

Each actuator class is mounted to its own physically identical thrust block, restrained from a fixed base by three strain-gauge load cells acting through flexures, one axial and two lateral, resolving the complete thrust vector in both magnitude and direction. The cells are sized to the expected loading envelope of the test motor, using a five-kilogram axial cell and two one-kilogram lateral cells digitized at eighty samples per second, identically on both stands. An alternative single-piece design, a cruciform flexure instrumented with one bridge per arm forming a unified three-axis force sensor, is held as a fallback configuration should the discrete three-cell assembly prove difficult to align. Running the identical instrumentation chain on two separate physical stands, rather than swapping actuators on one shared fixture, is what makes the magnetic-versus-servo comparison valid under nominally identical thrust conditions. Commanded-versus-measured deflection angle is logged across a series of step and ramp commands under identical thrust conditions on each stand to extract bandwidth, slew rate, step-response overshoot, steady-state error, and maximum sustained deflection for each actuator. Because each firing provides only a few seconds of control window, the independent command set for each actuator system is built up across multiple firings on its respective stand.

The static-fire stand is a single-axis, load-cell-only stand fitted with a steel blast deflector. It validates the as-fired thrust curve of every motor class used in the program against its published specification, and carries a thermocouple on the engine-bay wall to measure the peak temperature that the materials heat-deflection argument depends on. The blast-shield materials screen runs on this same stand: a flat coupon plate of each candidate material, five millimetres thick at full density, is mounted directly in the exhaust path and fired on. Six materials go through the screen, and the measured response is melt-through, ablation depth, and slag buildup rather than thrust or deflection. Materials surviving five millimetres are retested at four, then three, then two millimetres until they fail, feeding the materials dataset as a failure-thickness ranking. Ground firings use a plugged, zero-delay variant of the flight motor, which has an identical thrust curve but no ejection charge, so that nothing fires into the stand fixtures after burnout.

The bench aerofoil rig provides direct lift, drag, and stall measurement for the four candidate fin sections across a half-degree angle-of-attack sweep, supplying the measured leg of both the fin-selection and the tunnel-versus-flight calibration questions.

Planned firing counts are four flight motors for flight testing, and between thirteen and twenty-four plugged ground motors: six across the two thrust-vector stands at three firings

per actuator system, two on the static-fire stand for thrust-curve verification and the engine-bay wall temperature measurement, and the blast-shield screen, which is adaptive rather than fixed. Six materials at up to four thickness steps each tops out at twenty-four firings if every material survives to two millimetres, though most candidates are expected to fail by four millimetres in practice, so the realistic budget is roughly twelve to sixteen firings with twenty-four as the worst case. A further six low-cost motors are budgeted for stand commissioning at two firings per motor-fired stand.

Each stand's load cells are first calibrated independently of any motor firing, using a series of known hanging dead weights spanning the expected force range, to establish a force-to-voltage transfer function for every channel. Each stand is then commissioned with a minimum of two low-cost motor firings before any data-collection firing is conducted, to validate the as-built stand's measured thrust curve against the motor's independently published reference curve and confirm that structural compliance in the stand itself is not corrupting the force measurement. Only after a stand passes this commissioning check are data-collection firings conducted on it. All raw and reduced ground-test data are archived in the same repository used for flight data release.

Simulation Suite and Software-in-the-Loop Validation

The computational tier carries the analytical and inviscid-computational load for aerodynamics and control, and is held to an explicit verification standard: every simulated result reported here is produced by a version-controlled script, paired with the dataset it wrote, and gated against hard pass/fail criteria rather than inspected qualitatively. Where a physical measurement exists for a given quantity, the simulated result is reported alongside it as a cross-check rather than as the only available answer.

A fourth-order Runge-Kutta integrator advances the two-dimensional point-mass state under the digitized thrust curve, the componentwise drag buildup described above, an atmosphere referenced to the measured surface temperature and pressure, and a power-law wind-shear profile. Monte Carlo dispersion samples the full field envelope, including mean wind, turbulence intensity, surface temperature and pressure, launch-rod angle, and site elevation, together with vehicle-side dispersions in liftoff mass, center-of-gravity station, drag coefficient, and total impulse, producing distributions of apogee, maximum dynamic pressure, deployment velocity, landing dispersion, and static margin at every point in the burn.

The pitch-plane control model reproduces the flight law as executed: the discrete controller at the firmware's rate, a second-order servo model with a slew-rate limit, an explicit control-loop transport delay, gimbal-rate and travel limits, and gust forcing generated from a turbulence spectrum rather than a single sinusoid. Frequency-domain phase and gain margins are evaluated at every point in a burn-time by atmosphere grid, and a candidate gain set is accepted only if it clears the margin target at every point in that grid.

The flight firmware's state machine, sensor models, and control law are exercised end-to-end against simulated barometer, inertial, and battery signals carrying representative noise, bias, and quantization, producing synthetic flight logs in the same schema as the onboard recorder. This lets the ground-station analysis pipeline and every pass/fail gate be

exercised and debugged before any motor is fired, and provides the second method for the actuator research question by driving the bench actuator models under the identical control law used on the balance.

The instrumented stands are additionally modelled at the signal level, covering load-cell full scale and bridge noise, amplifier sample rate and quantization, stand structural compliance and its resulting mount resonance, and actuator lag and linkage ratio, so that the expected measurement, its uncertainty, and the required cell sizing are predicted before the stand is built, and any departure of the as-built stand from that prediction is itself detectable.

All simulated aerodynamic, trajectory, and control-loop datasets generated during the program are version-controlled alongside the physical test data in the program's public repository, with each simulation script paired against the dataset it produced, so that every plotted or tabulated simulated result in this paper and its supporting materials can be regenerated and independently checked against the as-built hardware.

Flight Test Plan, Logging, and Data Sharing

Each flight is conducted in compliance with the National Association of Rocketry (2023) safety code and motor classification standards, under the supervision of a certified or similarly qualified range safety officer. Multiple flights are conducted with the control law configured under each candidate gain set, and post-flight telemetry recovered from the onboard log is analysed for peak pitch deviation during the powered phase, gimbal-angle utilization, and settling and overshoot behaviour, to determine which gain configuration produces the most accurate and best-damped attitude tracking. All onboard sensor data, including full-rate inertial, barometric, and control-loop telemetry, is logged at the full control-loop sample rate. Where a bench or range wireless link is in range, a parallel live feed is monitored for real-time anomaly detection, though the onboard log remains the data of record. Upon conclusion of the program, all design files, firmware source code, simulation scripts, and the complete flight and ground-test datasets are released through a version-controlled, publicly accessible repository, and the principal reduced results from each research question are additionally presented in summary form within this paper.

Recovery System

The vehicle's passive fin stability is retained through the coast phase to apogee, predicted near 409 feet at approximately 6.7 seconds after launch. Recovery is initiated not by an independent electronic altimeter but by the flight motor's own factory ejection charge. The flight configuration uses a motor with a four-second ejection delay, so that approximately four seconds after propellant burnout, roughly seven-tenths of a second past the predicted apogee, the motor's integral ejection charge fires inside the lower body tube. Gas pressure builds directly against the bulkhead joint between the two body tubes, and the joint, friction-fit and shear-pinned to release at a target force, separates, deploying the parachute packed at the forward end of the lower body tube.

A first-order feasibility analysis supports the approach on two independent grounds. Flow-path losses at the ejection mass-flow rate are negligible against the driving pressure. The

recovery bay pressurizes to roughly an order of magnitude above the friction-fit joint-release threshold, giving a pressurization margin of several times. The four-second delay is the closest available factory delay to the coast-to-apogee optimum; longer available delays are rejected because they fire well past apogee, deploying at high descent speed and, in the longest case, at dangerously low altitude.

This motor-ejection architecture needs no independent recovery computer, no isolated recovery battery, no electric match or black-powder charge, and no dedicated recovery wiring, reducing parts count, cost, and an entire electronic failure domain, at the cost of a single passive deployment event with no electronic backup channel. That trade is justified by the pressurization margin and by the finned airframe's aerodynamic stability through apogee. Shock-cord and parachute sizing, using tubular aramid cord and a 24-inch ripstop nylon canopy with an aramid blanket shielding the canopy from the ejection gas, are verified against the worst-case deployment scenario, giving a harness structural safety factor comfortably above target. A predicted terminal descent rate near five metres per second follows once the canopy is fully open.

All electrical connections crossing the bulkhead, the two servo leads and the gimbal inertial unit cable, are made on separable male-female extension leads that simply pull apart at separation, so the aramid shock cord is the only retained link between the two halves. The firmware treats the resulting loss of the gimbal sensor and servo power as an expected consequence of deployment rather than as a fault condition.

Safety and Regulatory Compliance

All flights use a single Estes F15-4 motor, an F-class motor with a four-second ejection delay, and the fully loaded liftoff mass of approximately 720 grams is well under the Federal Aviation Administration's Class 1 model rocket threshold of 1,500 grams loaded weight per motor, requiring no airworthiness waiver and no high-power certification. Range procedures include remote ignition, a minimum three-metre personnel standoff from both ground-test stands during firing, a fail-safe neutral-gimbal default state on any control-system fault, and standard model-rocketry motor-handling discipline for the motor's integral ejection charge. The igniter is installed last, at the pad, and there are no independent pyrotechnic or electronic ejection circuits in the vehicle to arm or inhibit, since recovery is effected solely by the motor's own delay and ejection charge.

Expected Outcomes

The program is expected to produce a quantitative, ground-validated comparison of magnetic-solenoid and servo thrust-vector-control actuators, including bandwidth, slew rate, overshoot, and steady-state error for each, directly informing actuator selection for future closed-loop rocketry programs without requiring a dedicated in-flight comparison.

A validated thermally-zoned additive-manufacturing material allocation is expected to demonstrate a measurable dry-mass reduction relative to a uniform heat-rated-material baseline, with no corresponding loss of structural margin, offering a transferable design pattern for hobbyist and academic rocketry programs using fused-deposition fabrication.

The blast-shield screen adds a failure-thickness ranking across six candidate materials under real motor-plume conditions to that same dataset.

Direct wind-tunnel measurement of the candidate fin aerofoil sections, cross-checked against the inviscid computational prediction, is expected to produce a defensible, measured basis for fin-section selection rather than a purely modelled one, and, combined with flight telemetry, a quantified accuracy bound on Barrowman-class stability prediction for short, low-Reynolds-number, additively-manufactured airframes, expressed as the discrepancy between predicted, tunnel-measured, and telemetry-reconstructed static margin and coast-phase drag coefficient. This three-way comparison will be released as an open dataset, giving subsequent low-cost programs a defensible error bar to design against with or without their own flow facility.

Finally, flight-validated comparison of control-loop gain configurations on a consolidated dual-core single-controller architecture is expected to clarify the practical control-authority and determinism benefits of separating real-time control execution from logging and telemetry input/output on a single low-cost microcontroller, relative to either a single-threaded controller or a distributed multi-board avionics stack. All resulting design files, firmware, simulation code, and flight datasets will be released publicly in support of the program's open-source objectives.

Project Timeline

The build-to-flight schedule runs on a fixed external constraint: a primary launch window on the weekend of Saturday and Sunday, November 7 and 8, 2026, with a contingency window the following weekend, November 14 and 15, 2026, held in reserve for weather scrubs, hardware anomalies, or a repeat flight if the primary window leaves a gap in the dataset. All program data collection concludes by December 1, 2026, with the intervening weeks held as unscheduled margin rather than committed work. Work is budgeted at three hours per week against this fixed end date, excluding unattended print run time but including computer-aided design, print-queue management, and all post-processing.

The schedule proceeds through design lock and procurement; electronics bring-up on the avionics stack; design and fabrication of all four ground-test stands; the ground-test campaigns for the actuator, materials, and aerofoil questions together with the motor-characterization data supporting the control-gain question; airframe fabrication and recovery-system integration; full-stack integration and a dry-run countdown rehearsal; the two launch windows; and a final data-reduction pass cross-checking results against all five research questions. The governing principle is that the launch dates are fixed and the schedule bends around them, with any slip absorbed by the built-in end-of-schedule buffer before either launch date is moved.

References

Abbott, I. H., & Von Doenhoff, A. E. (1959). *Theory of wing sections: Including a summary of airfoil data*. Dover Publications.

Barrowman, J. S. (1967). *The practical calculation of the aerodynamic characteristics of slender finned vehicles* (NASA NTRS accession 20010047838).
<https://ntrs.nasa.gov/citations/20010047838>

Bosch Sensortec. (n.d.). *BNO085 9-axis absolute orientation IMU, datasheet*.

BPS.space. (n.d.). *Thrust vector control*. <https://bps.space/products/thrust-vector-control>

Dizon, J. R. C., Espera, A. H., Chen, Q., & Advincula, R. C. (2018). Mechanical characterization of 3D-printed polymers. *Additive Manufacturing*, 20, 44–67.
<https://doi.org/10.1016/j.addma.2017.12.002>

Estes Industries. (n.d.). *F15 model rocket engine, technical specifications*.

Federal Aviation Administration. (n.d.). *Title 14 CFR Part 101, Subpart C, Amateur rockets*.

Kuethe, A. M., & Chow, C.-Y. (1998). *Foundations of aerodynamics: Bases of aerodynamic design* (5th ed.). John Wiley & Sons.

Lissaman, P. B. S. (1983). Low-Reynolds-number airfoils. *Annual Review of Fluid Mechanics*, 15, 223–239. <https://doi.org/10.1146/annurev.fl.15.010183.001255>

Mueller, T. J., & DeLaurier, J. D. (2003). Aerodynamics of small vehicles. *Annual Review of Fluid Mechanics*, 35, 89–111. <https://doi.org/10.1146/annurev.fluid.35.101101.161102>

National Association of Rocketry. (2023). *Model rocket safety code*.
<https://www.nar.org/safety-information/model-rocket-safety-code/>

Popescu, D., Zapciu, A., Amza, C., Baci, F., & Marinescu, R. (2018). FDM process parameters influence over the mechanical properties of polymer specimens: A review. *Polymer Testing*, 69, 157–166. <https://doi.org/10.1016/j.polymertesting.2018.05.020>

Raspberry Pi Ltd. (2024). *Raspberry Pi Pico 2 W datasheet*.